

Digital metadata of tangible and intangible heritage for Polyvocal Cultural AI Practices

Victor de Boer
and Cultural AI lab collaborators

A young woman with long brown hair in a braid, wearing black-rimmed glasses and a white t-shirt, is smiling while working on a laptop. She is in a modern, creative office environment with various objects like pens, a lamp, and a desk organizer visible in the background.

AI in the Cultural Heritage sector:
Recommender systems
Automatic classification
Tagging, metadata creation and
enrichment

A woman with long brown hair in a braid, wearing black-rimmed glasses and a white t-shirt, sits at a desk. She has her hand to her chin in a thoughtful pose, looking at a laptop screen. The desk is cluttered with various items, including pens, pencils, and a container. The background is a blurred office environment.

Transparency?
Privacy?
Inclusivity?
Diversity?



cultural-ai.nl

Cultural AI Lab



ELSA Lab

ICAI Innovation Center for Artificial Intelligence



RIJKS MUSEUM





Steven Claeysens @sclaeysens · Oct 12

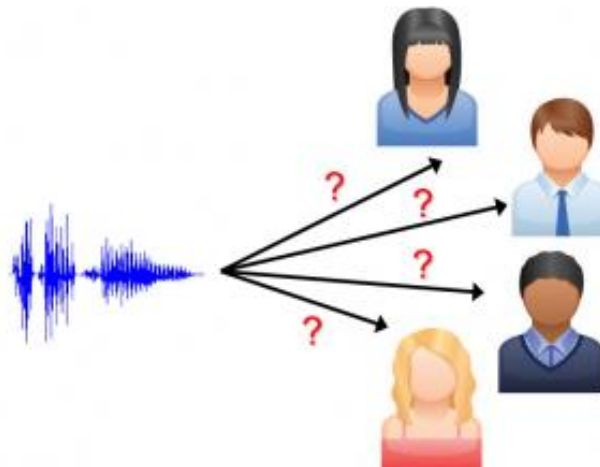
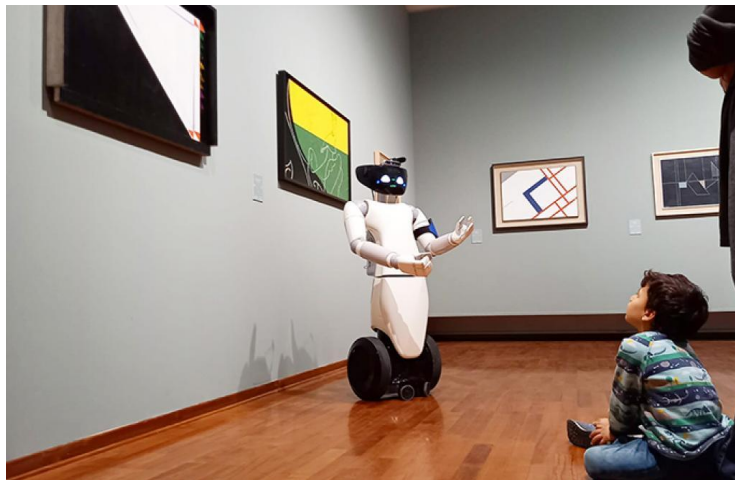
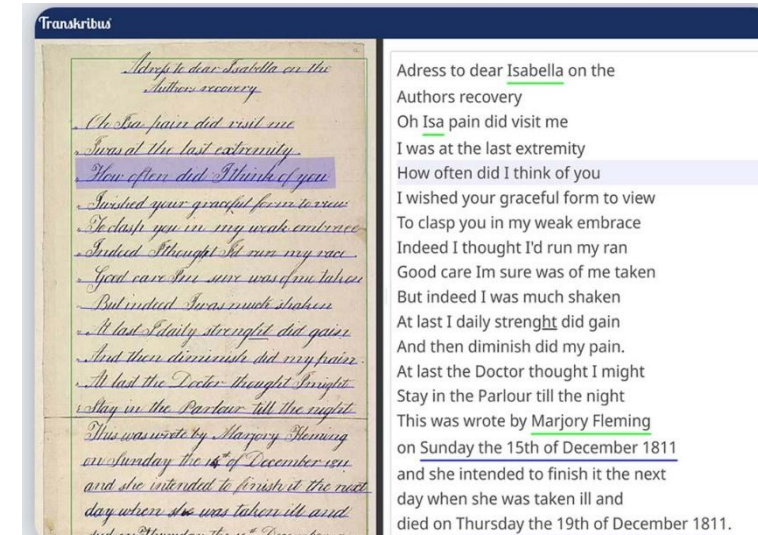
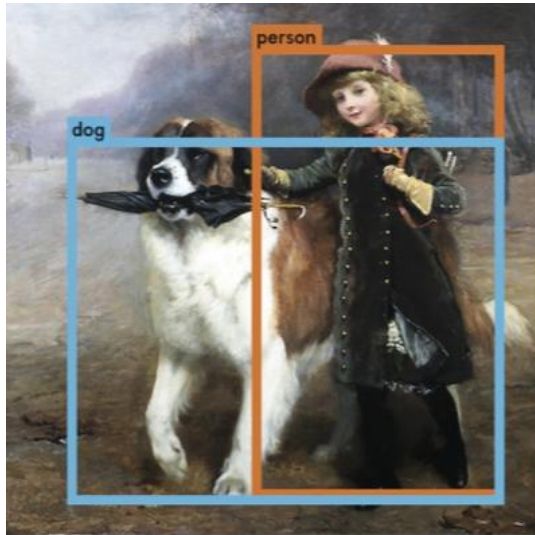
At last, an opportunity to repost this meme. 'AI also has a lot to learn from culture'. Cultural AI Lab introduced by [@victordeboer](#)
[#EuropeanaTech2023](#)
[cultural-ai.nl](#)

The image is a meme featuring two men. On the left, a man is wearing a very elaborate and voluminous pink feathered costume, a blue and white headpiece, and large, white-rimmed sunglasses. He has a surprised or expressive facial expression. On the right, a man with dark hair, wearing a dark jacket, is looking towards the person in the costume with a neutral, attentive expression. The background is slightly blurred, showing what appears to be an indoor setting with some equipment.

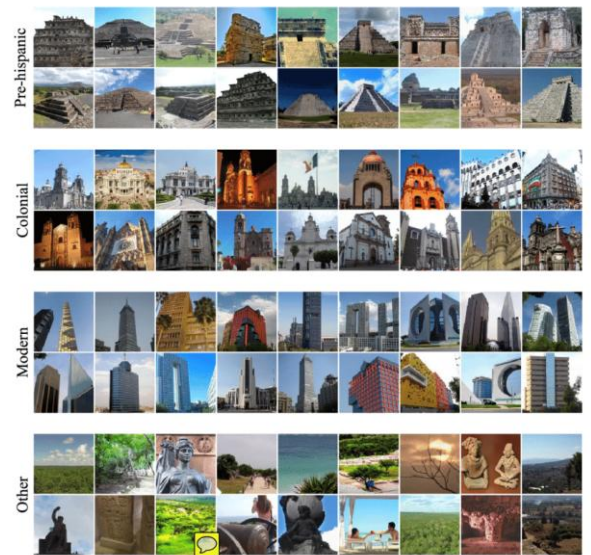
Culture for AI

AI for Culture

AI for Culture



Culture for AI



A screenshot of the Transkribus interface. On the left, a handwritten letter is shown. On the right, the transcription is displayed with the following text:

Adress to dear Isabella on the Authors recovery
Oh Isa pain did visit me
I was at the last extremity
How often did I think of you
I wished your graceful form to view
To clasp you in my weak embrace
Indeed I thought I'd run my ran
Good care Im sure was of me taken
But indeed I was much shaken
At last I daily strenght did gain
And then diminish did my pain.
At last the Doctor thought I might
Stay in the Parlour till the night
This was wrote by Marjory Fleming
on Sunday the 15th of December 1811
and she intended to finish it the next
day when she was taken ill and
died on Thursday the 19th of December 1811.



A diagram illustrating the process of mitigating biases in AI. It shows a flow from 'Shoes' (represented by an icon of a shoe) to 'Standard English' (represented by the text 'Standard English'). This leads to a robot head icon. A speech bubble from the robot says 'Singlish ??? "I will handle the bookings. Boleh?"'. Another speech bubble shows a high-heeled shoe icon and '???'.

Mitigating Biases:

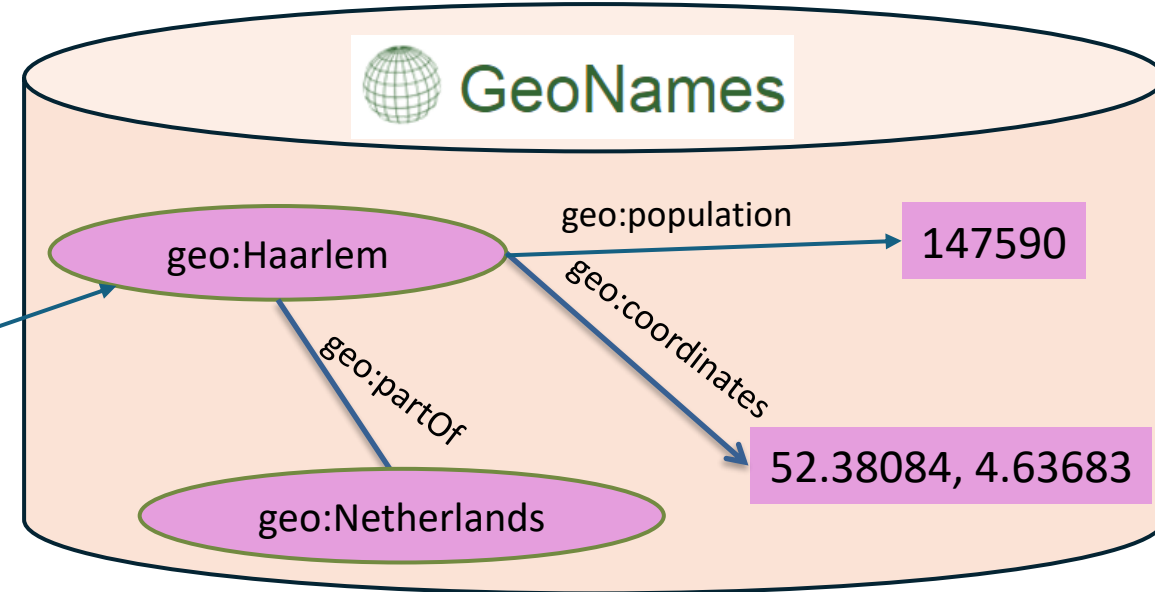
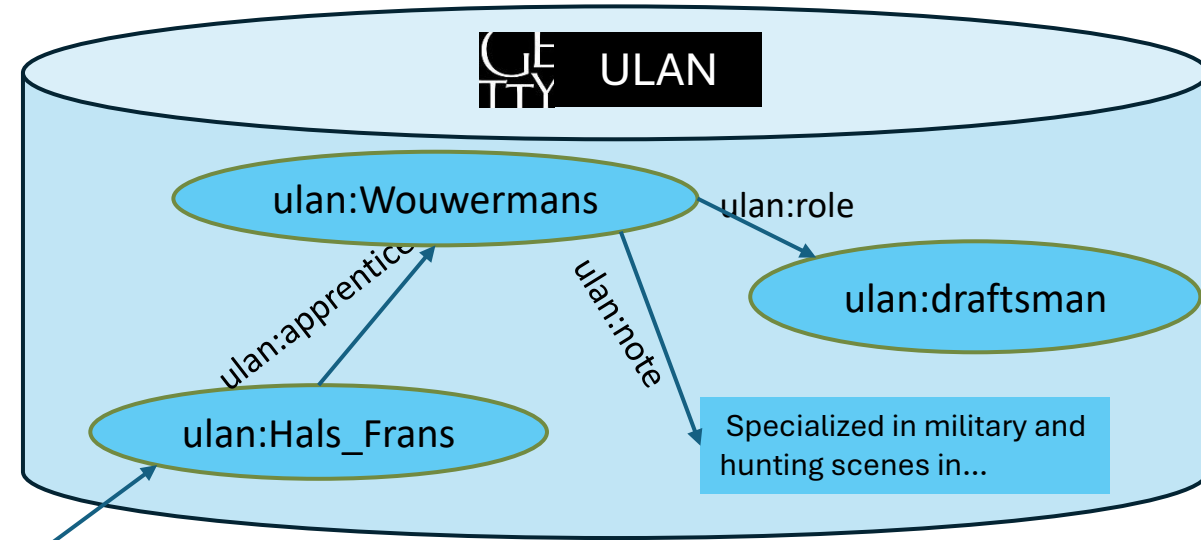
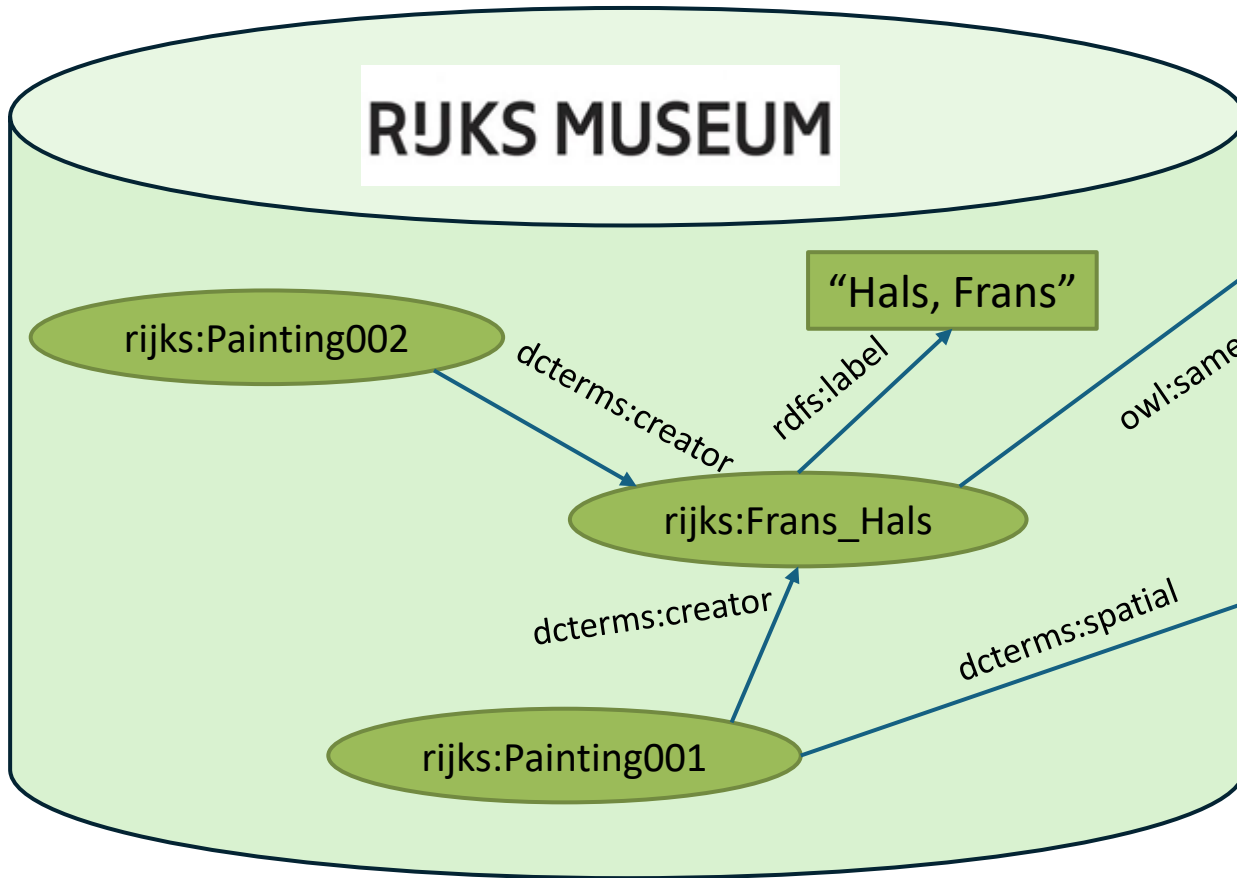
Standard English

Building Inclusive AI Products

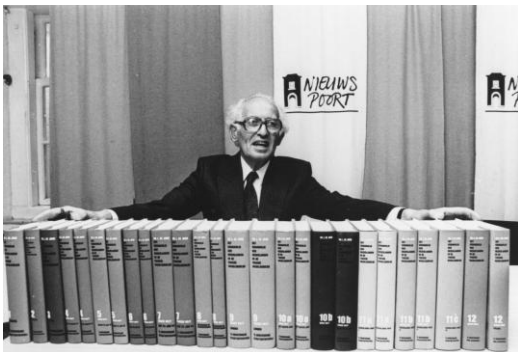
How to connect heritage in an
inclusive, polyvocal manner?



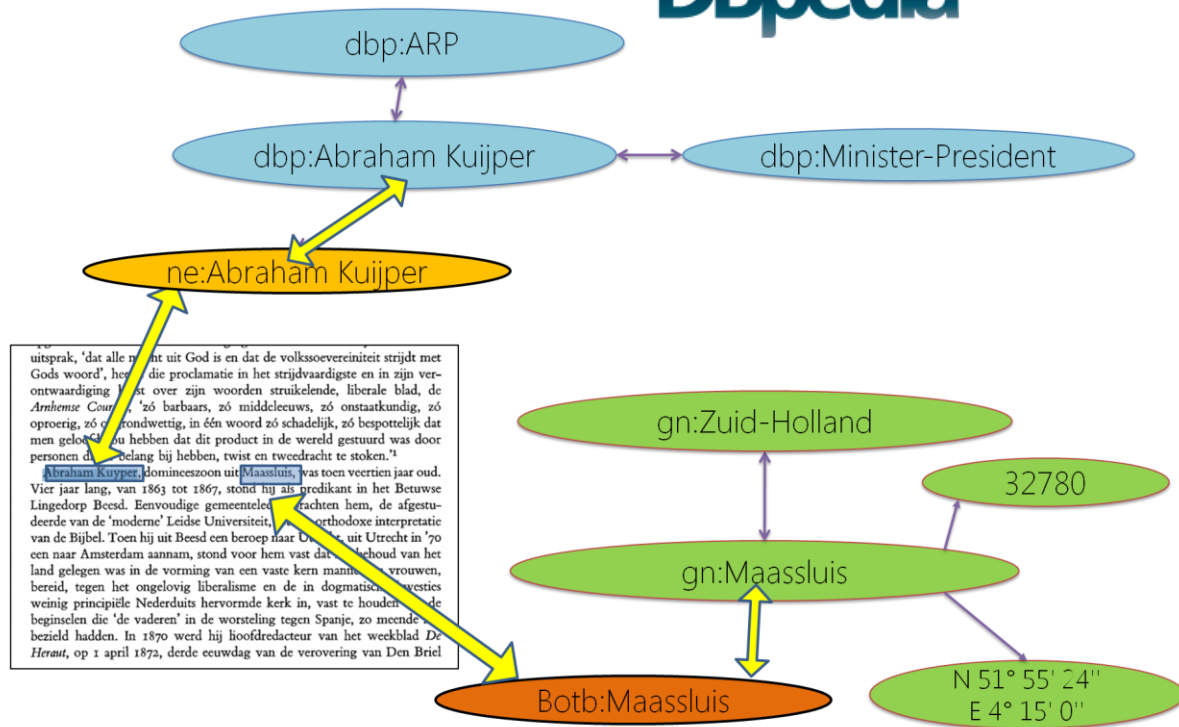
Web of Linked Data consisting of connected Knowledge Graphs



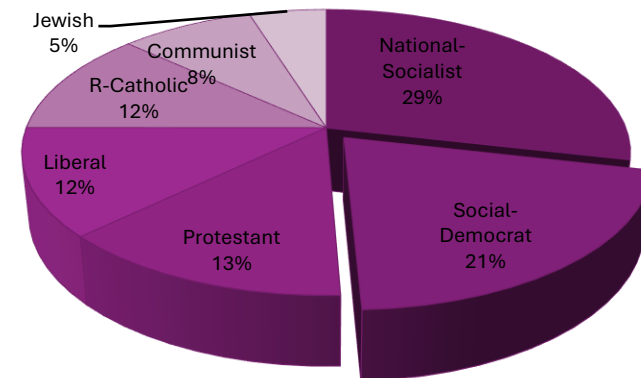
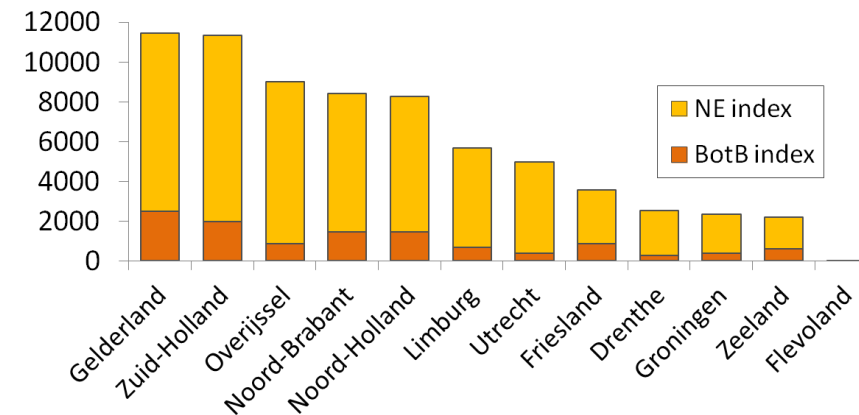
owl:sameAs



Analyses and visualisation of knowledge graphs



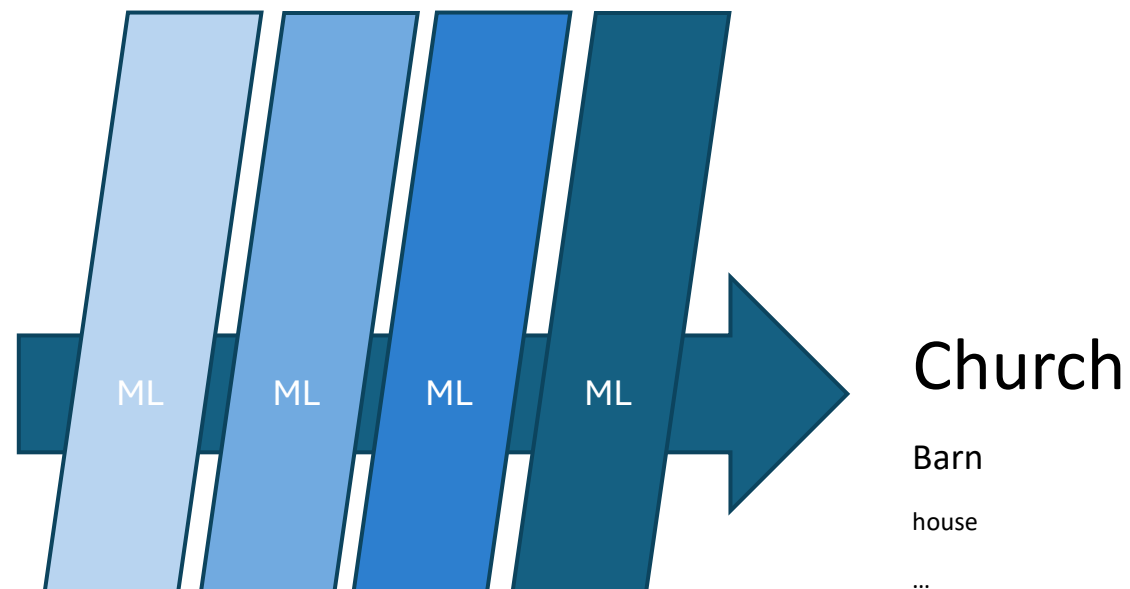
Example enrichment: through links with GeoNames
(via Back of the Book index) and DBpedia (via NE index)



Machine learning on knowledge graphs



WebPage		0 ERRORS 0 WARNINGS ^
@type	WebPage	
url	https://data.labs.kadaster.nl/kadaster/bag-vocab/browser?resource=https%3A%2F%2Fdata.labs.kadaster.nl%2Fkadaster%2Fbag-vocab%2Fdef%2FPand	
mainEntity		
@type	Class	
@id	https://data.labs.kadaster.nl/kadaster/bag-vocab/def/Pand	
description	Een pand is een kleinste bij de totstandkoming functioneel en bouwkundig-constructief zelfstandige eenheid die direct en duurzaam met de aarde is verbonden en betreedbaar en afsluitbaar is	
name	pand	
url	https://data.labs.kadaster.nl/kadaster/bag-vocab/browser?resource=https%3A%2F%2Fdata.labs.kadaster.nl%2Fkadaster%2Fbag-vocab%2Fdef%2FPand	



Based
on what
data?

Which
voices
are we
missing?



“Ziggurat of selection”



Knowledge
Graph
construction

Digitization

Selection

Archiving

Survival

Social
Selection

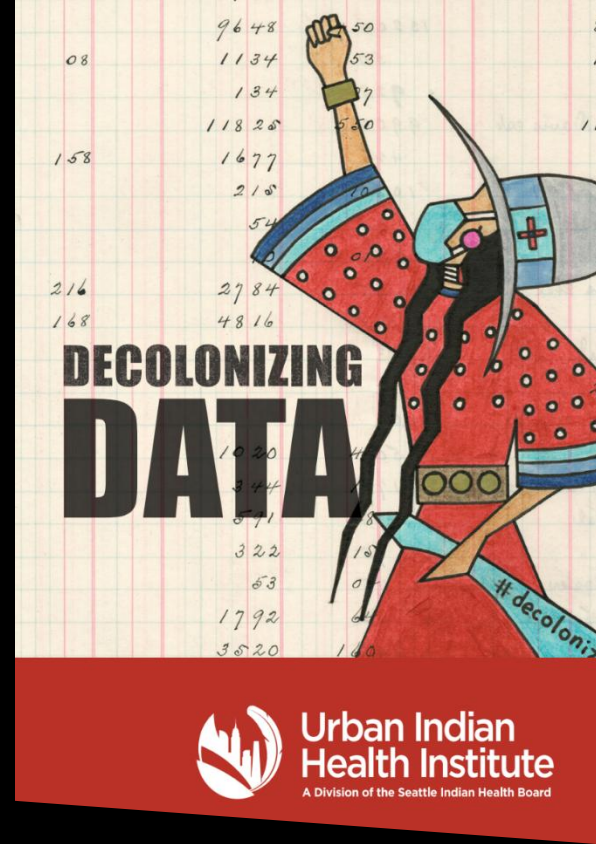
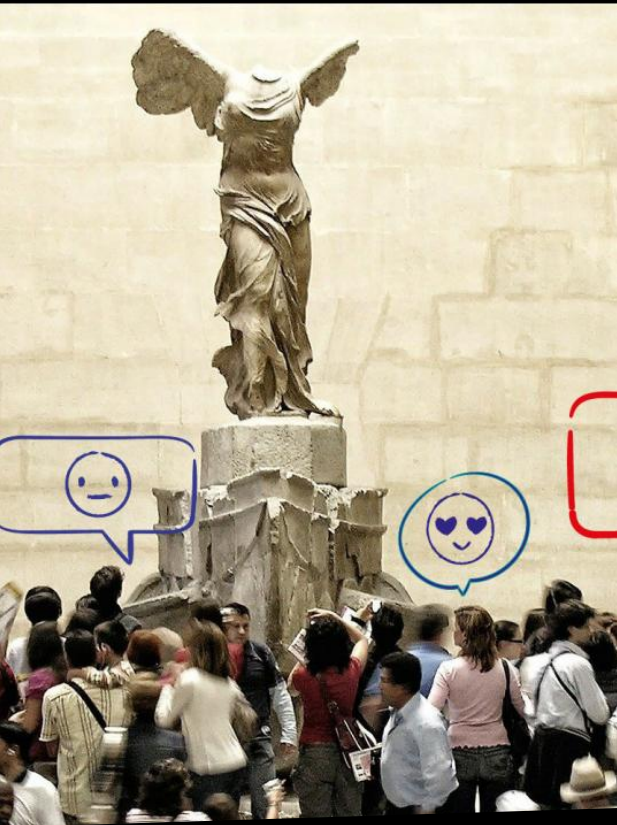


Polyvocality

Polyvocal, *adjective*: In narrative strategy, an approach that relies on the power of many voices speaking the same core change.

<https://narrativeinitiative.org/blog/polyvocal-narrative-strategy-turning-many-voices-into-durable-change/>,

choir



Polyvocality and decolonization in museums, archives, libraries



Knowledge Graphs for Polyvocality

● Polyvocality for Knowledge Graphs

Identifying and acquiring polyvocal knowledge

Representing of polyvocality: models, patterns

Presenting of polyvocal knowledge to variety of users

A Polyvocal and Contextualised Semantic Web*

Marieke van Erp¹[0000-0001-9195-8203] and Victor de Boer²[0000-0001-9079-039X]

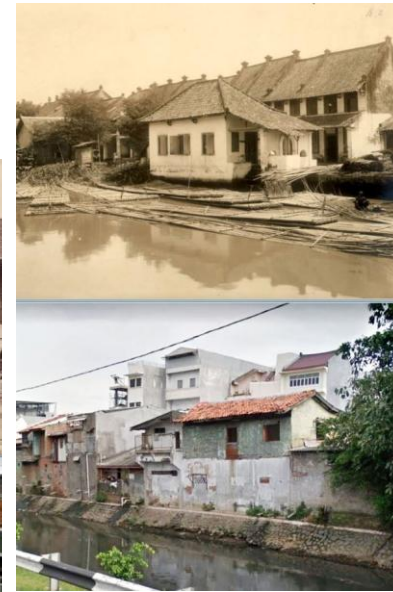
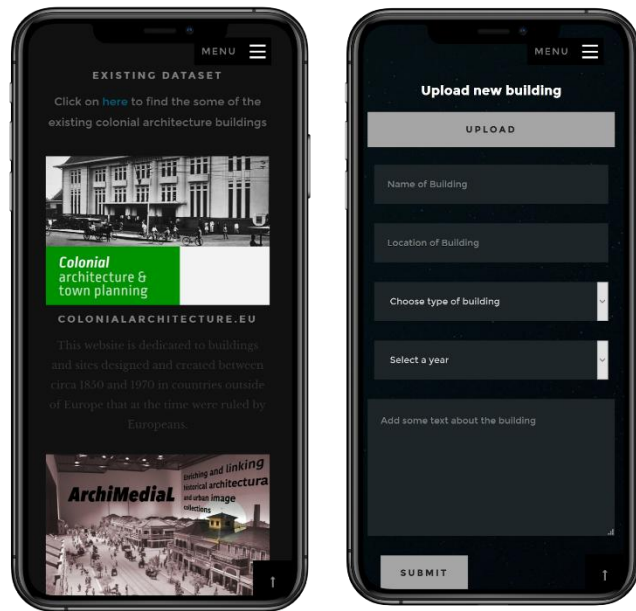
¹ KNAW Humanities Cluster
Amsterdam, the Netherlands
marieke.van.erp@huc.knaw.nl
<https://mariekervanerp.nl>

²

Identifying and
acquiring polyvocal
knowledge



Crowdsourcing perspectives for colonial architecture



Roz Sabir

Nichesourcing for polyvocal CH object annotation



World Museum objects
from Kuching/Sarawak

1. What is this object (*class*)
2. What is the purpose of this object (*usage*)
3. How should the object be treated (*care*)



Lea Krause

Representing
polyvocal
knowledge



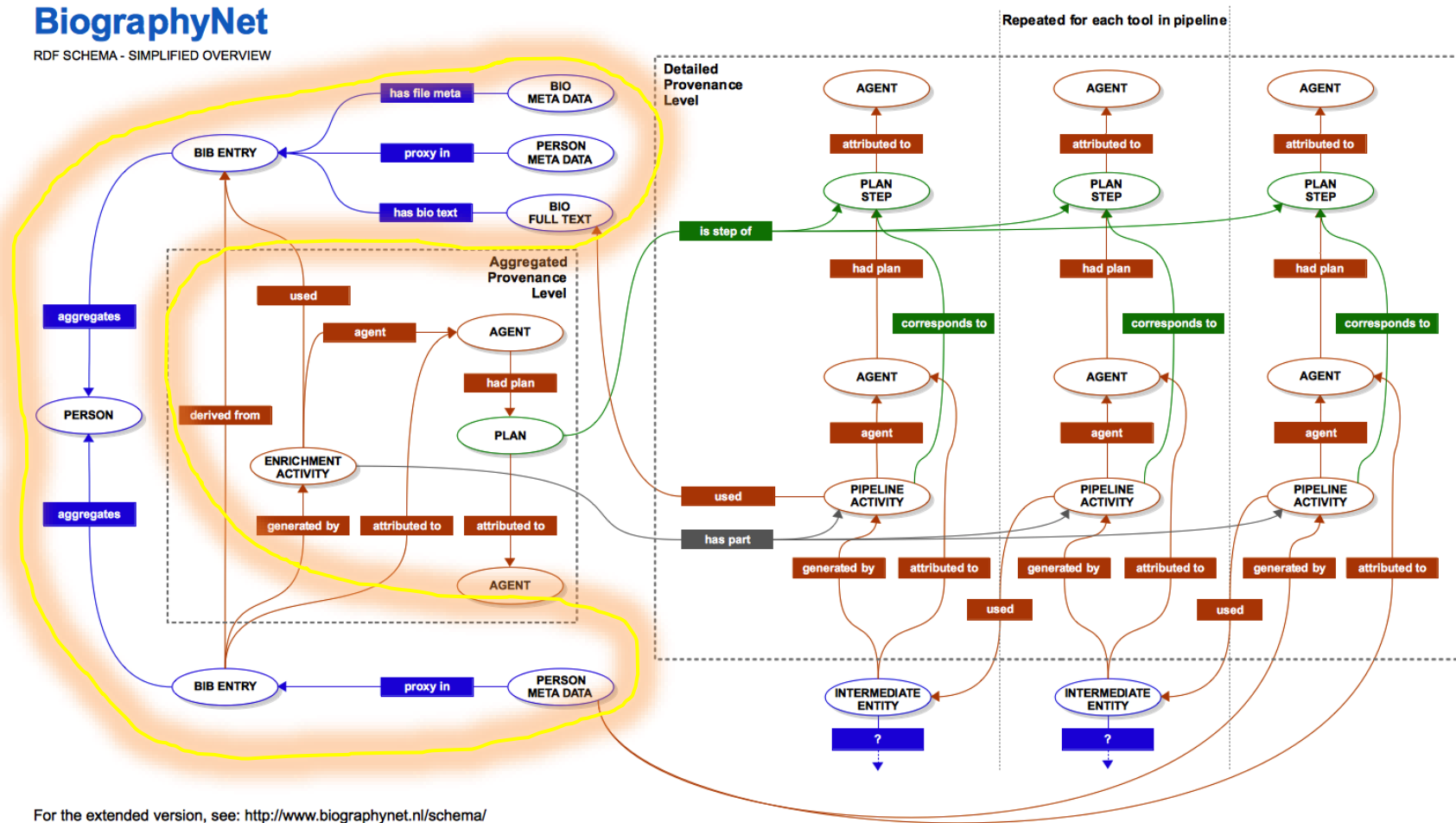
Polyvocality and cultural context in Biographical knowledge graphs



Niels Ockeloen



Go Sugimoto



Rijksuniversiteit Groningen: Roofkunst moet terug, maar 'mag niet in een schuur'

NET 8

Berlin praised for returning Benin bronzes



The artifacts feature animal and human figures and were created from the 16th century onwards. ALAMY

Germany has been hailed for leading a reappraisal of Europe's colonial past after Berlin announced the return of priceless [Benin bronzes](#) to Africa.

The move increases pressure on Britain to repatriate treasures looted in the sacking of Benin City in 1897, when it was part of the Kingdom of Benin. Today the city is in Edo state, Nigeria, where government officials welcomed Germany's decision and called for Britain to follow its lead.



Advies: koloniale roofkunst moet onvoorwaardelijk terug

Levenswijzen en cultuur van volken uit andere landen zijn vaak vernietigd of vergeten. Het is tijd om deze kunstwerken terug te geven.

Wat het om gaat is schoon schip maken met het verleden

De Nederlandse overheid moet zich aanpakken op het gebied van koloniale roofkunst.

8°C 0 km

'MUSEUM NIET LEEG'



door Olav Verburg

Een van de meest gevoelige vragen die de museumwereld tegenwoordig stelt, is hoe we omgaan met de koloniale roofkunst die in onze musea staat. Het is een vraag die niet alleen over kunst gaat, maar ook over ethiek en recht.

Ik heb vaak geleerd dat de koloniale roofkunst een belangrijk onderdeel is van de Nederlandse cultuur. Het is een deel van ons verleden dat we niet mogen vergeten.

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Koloniale kunst Terug naar de eigenaar

'Behandel alle landen hetzelfde'

Veel bijl. oogsten de aanbestedingen over hoe Nederland moet omgaan met kunstobjecten uit de koloniale tijd. Maar kritiek is er ook.

door Henk van der Meer

1000 artikelen uit het verleden

LANDSLIP BEZORGEN

NIEUWS & ACHTERGROND • GELOOF & KERK • OPINIE & COLUMNS • MENSEN

Musea staan achter advies om koloniale roofkunst terug te geven aan landen van herkomst

Columns & Opinion Uitgelicht Wetenschap Mensen De Gids Cultuur & Media

NIEUWS KOLONIALE ROOFKUNST

Musea staan achter advies teruggave koloniale roofkunst aan landen van herkomst – maar niet

Netherlands

Dutch museums vow to return art looted by colonialists

Estimated 100,000 exhibits may be repatriated to remedy historical 'injustices'



The Great Hall of the Tropenmuseum, which houses collections about world cultures, in Amsterdam. Shutterstock

Advies van minister: koloniale roofkunst onvoorwaardelijk teruggeven

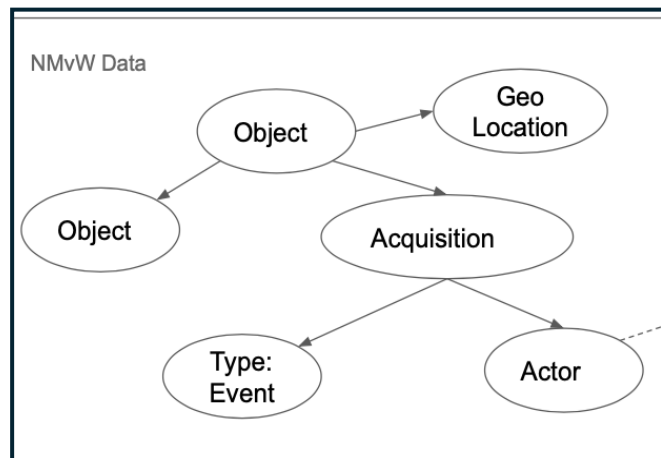
5 dagen geleden



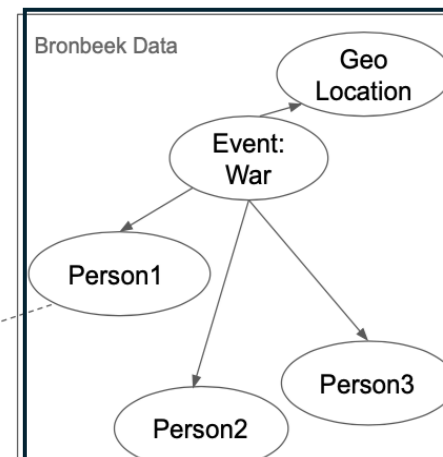
Knowledge Graphs support connected narratives



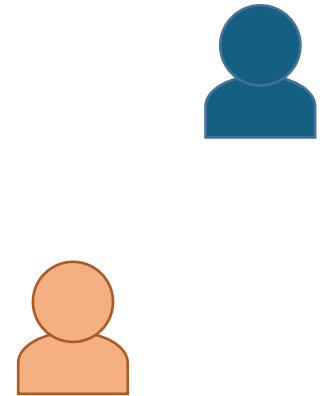
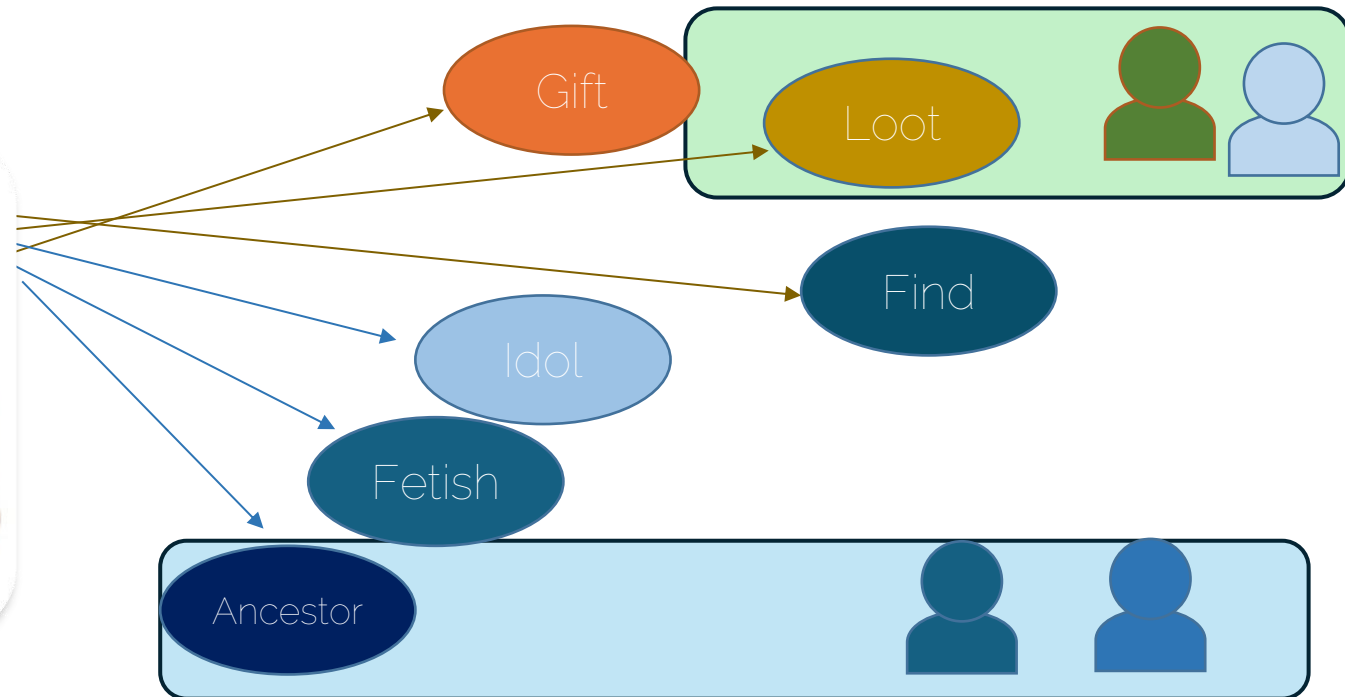
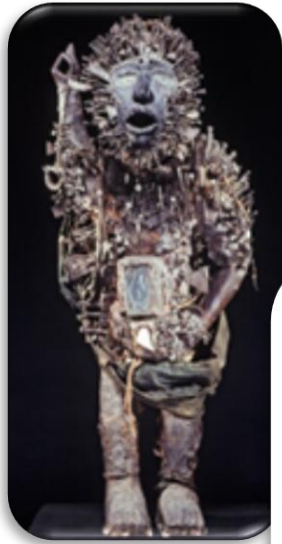
Wereldmuseum



Bronbeek museum



Sarah Shoilee



- Data-driven pattern analysis to challenge current metadata
- Allow community members to add their annotations (nanopublications)



Sarah Shoilee

data.colonialcollections.nl/data-hub/knowledge-graph

Colonial Collections Datahub

The Colonial Collections Datahub is a digital platform that brings together, enriches and provides insights into information on collections from colonial contexts. It includes heritage from Suriname, Indonesia, Sri Lanka, South Africa, Aruba, Curaçao, Sint Maarten and many other countries and areas. The datahub provides communities and researchers space to add perspectives on the included data. [See how this works](#)

Search for heritage objects

Tip: start searching for a location, e.g. Suriname



Add your narratives to heritage and objects

Your knowledge about the heritage on this platform is vital. With your narratives and perspectives, we can present the 'objects' in their context and point to biases in collection metadata.

When you create an account, you can add your narrative to these 'objects', make communities and create lists.

Create an account

Heritage communities

Upon creating an account, every individual or group of people can start their community on this platform. A community is able to create lists of objects, bringing together their heritage. Communities can also add narratives to the objects in these collections.

Title

SPEER

Provided by **Wereldmuseum**

A title or name given to the object by its current holder.

Add your narrative to title

Add a narrative

Narrative*

Here you can add your own narrative to the Description of this object.

Resource*

Please provide information that supports your narrative.

☐ I agree that all added information in the application is published with a [CC-BY-4.0 license](#)

This means that the information may be:

- copied and redistributed in any medium or format for any purpose, even commercially.

Language

Select a language of the narrative. Start typing to select a language.

English (English)

Community*

Select the community you speak on behalf of.

Select a community



License:
Attribution-ShareAlike 4.0 International (CC BY-SA 4.0)

Presenting
polyvocal
knowledge



Argumentation for **explainable** inconsistency resolving in **polyvocal** knowledge graphs



Loan Ho

Consider $\mathcal{K}_1 = \{\mathcal{R}_1, \mathcal{C}_1, \mathcal{F}_1\}$ where:

$$\mathcal{R}_1 = \{R : \forall x \text{Person}(x) \rightarrow \exists y \text{hasDeathdate}(x, y)\},$$

$$\mathcal{C}_1 = \{C : \forall x, y, z \text{Person}(x) \wedge \text{hasDeathdate}(x, y) \wedge \text{hasDeathdate}(x, z) \rightarrow y = z\},$$

$$\mathcal{F}_1 = \{f_1 : \text{Person}(\text{Thorbecke}), f_2 : \text{hasDeathdate}(\text{Thorbecke}, 14/10/1860),$$

$$f_3 : \text{hasDeathdate}(\text{Thorbecke}, 10/10/1860)\}$$

$$A_2 = (\{\text{Person}(\text{Thorbecke})\}, \{\text{hasDeathdate}(\text{Thorbecke}, 10/10/1860)\})$$

$$\uparrow \forall x, y, z \text{Person}(x) \wedge \text{hasDeathdate}(x, y) \wedge \text{hasDeathdate}(x, z) \rightarrow y = z$$

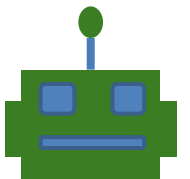
$$A_1 = (\{\text{Person}(\text{Thorbecke})\}, \{\text{hasDeathdate}(\text{Thorbecke}, 14/10/1860)\})$$



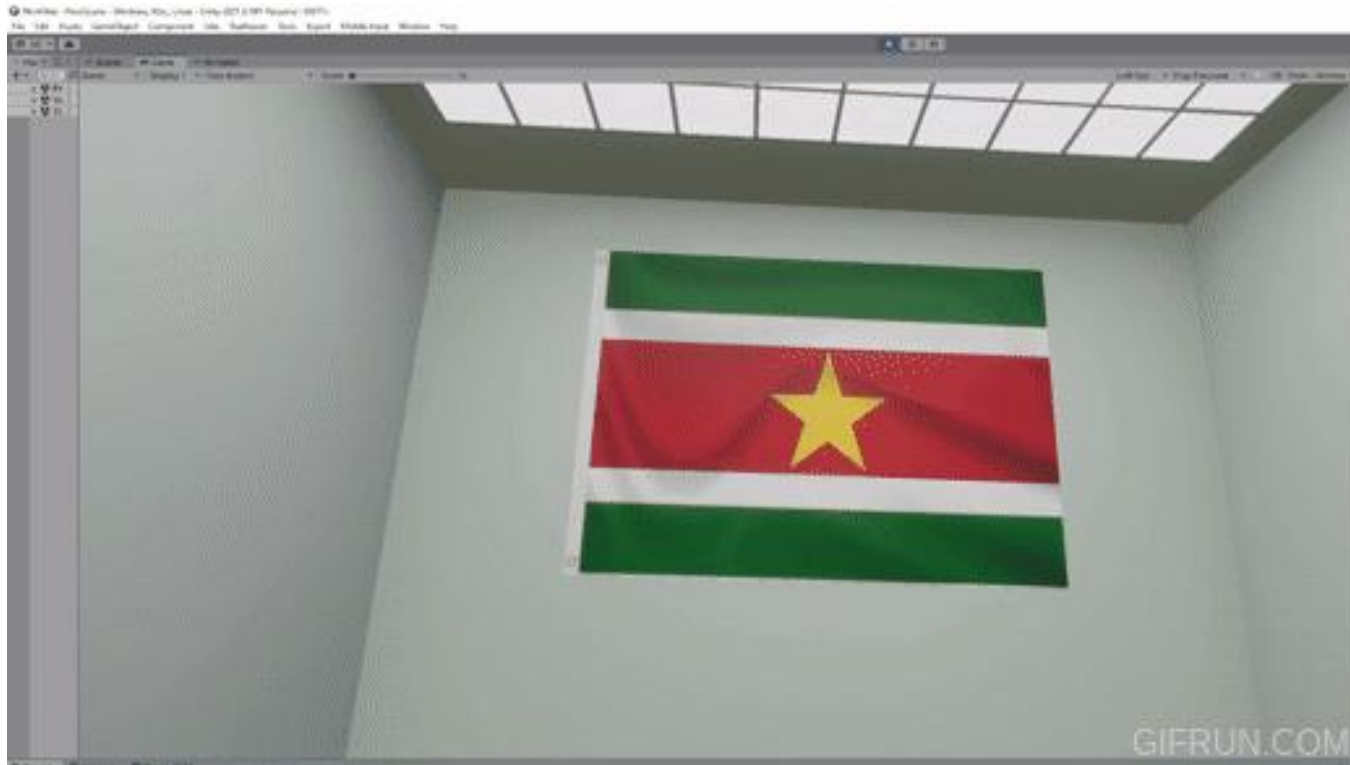
User: Why not
 $\text{hasDeathdate}(\text{Thorbecke}, 10/10/1860)$
given that A_2 ? ⁸

User: I understood "why 10/10/1860 is
not Thorbecke's death date"

Reasoner: Because
 $\text{hasDeathdate}(\text{Thorbecke}, 10/10/1860)$ ⁹
the following constraint is violated:
 $\forall x, y, z \text{Person}(x) \wedge \text{hasDeathdate}(x, y) \wedge$
 $\text{hasDeathdate}(x, z) \rightarrow y = z,$



Virtual Reality for Polyvocal exhibitions



Stefania Conte



Isabel Florence Franke



Claudia Libbi



Tilo Hartmann



Victor de Boer



HORIZON-CL2-2025-01-
HERITAGE-04

Through an in-depth investigation of AI applications in the Cultural and Creative Industries (CCIs), the HUMAIN project seeks to provide a clear understanding of both the opportunities and challenges facing the sector and strengthen a responsible use of technology.

**Navigating Human-Centred AI for Resilient and Sustainable
Cultural and Creative Ecosystems**



Thank you

Knowledge Graphs to connect polyvocal heritage

Challenges remain on

- how to get (to) these voices

- how to model their perspectives

- how to present these voices to variety of end-users

(Data) provenance is key

In the Cultural AI Lab, we investigate this in interdisciplinary teams

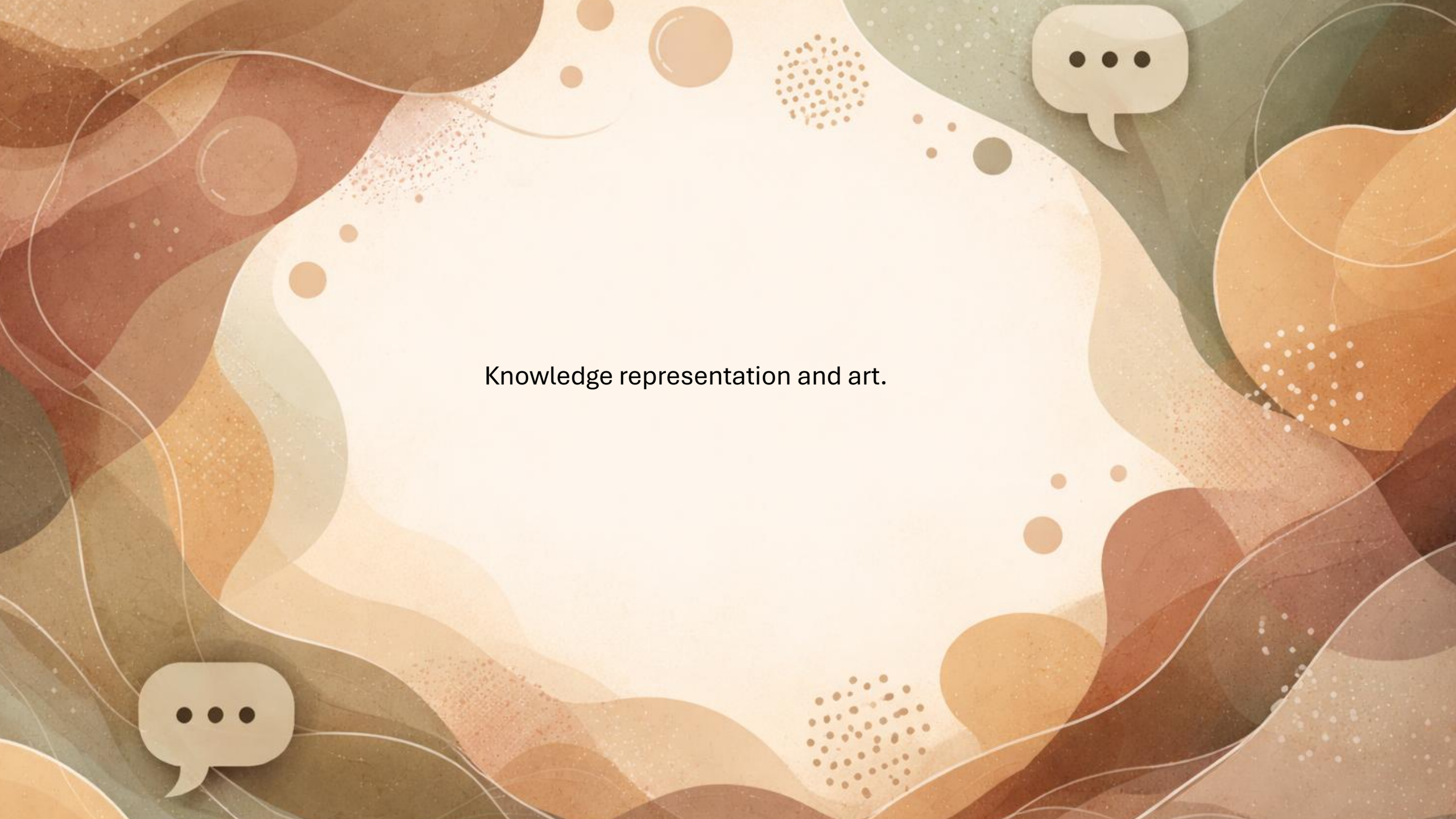
- We don't have all the answers

- We don't have all the perspectives

<http://www.victordeboer.com>

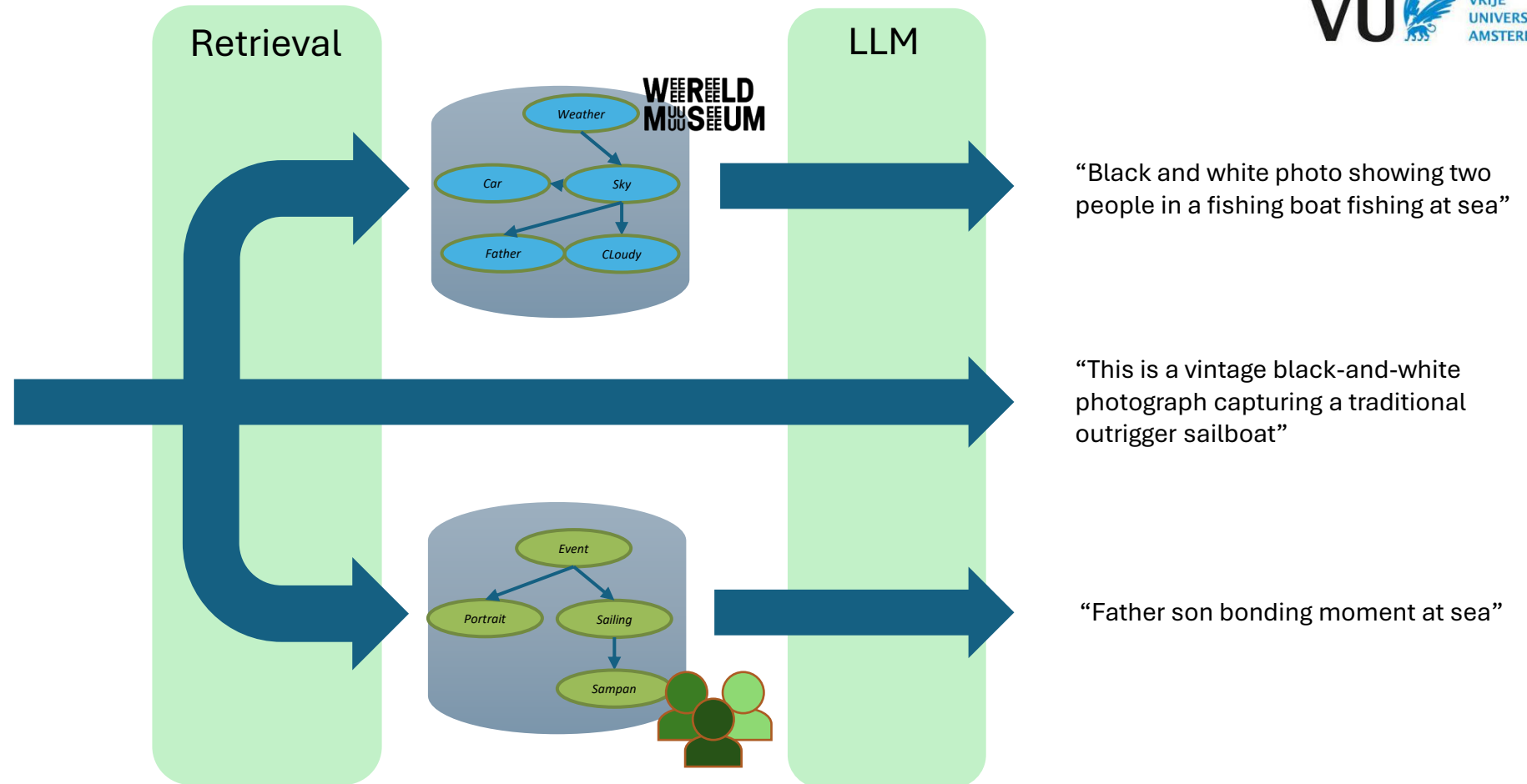
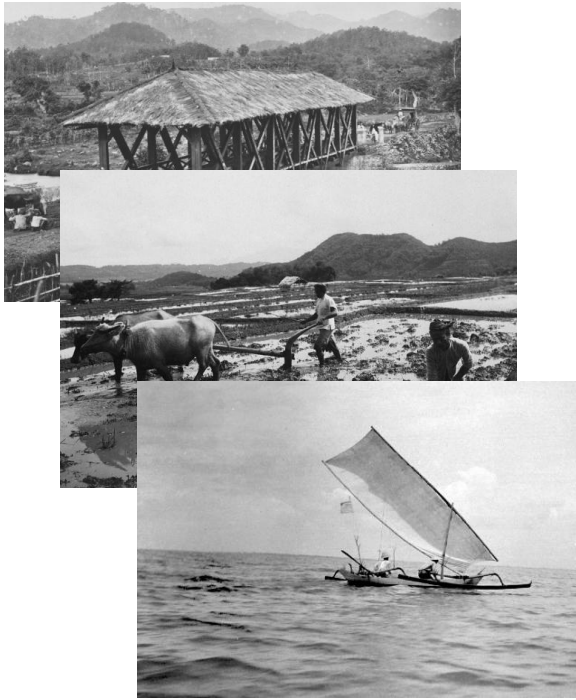
<https://www.cultural-ai.nl/>



The background is a complex, abstract composition of organic, flowing shapes in various shades of brown, tan, and olive green. These shapes overlap and blend into each other, creating a sense of depth and movement. Scattered throughout the composition are numerous small, solid circles and larger clusters of tiny dots, some in white and some in brown. Two light beige speech bubbles, each containing three black dots, are positioned in the upper right and lower left corners, pointing towards the central text area.

Knowledge representation and art.

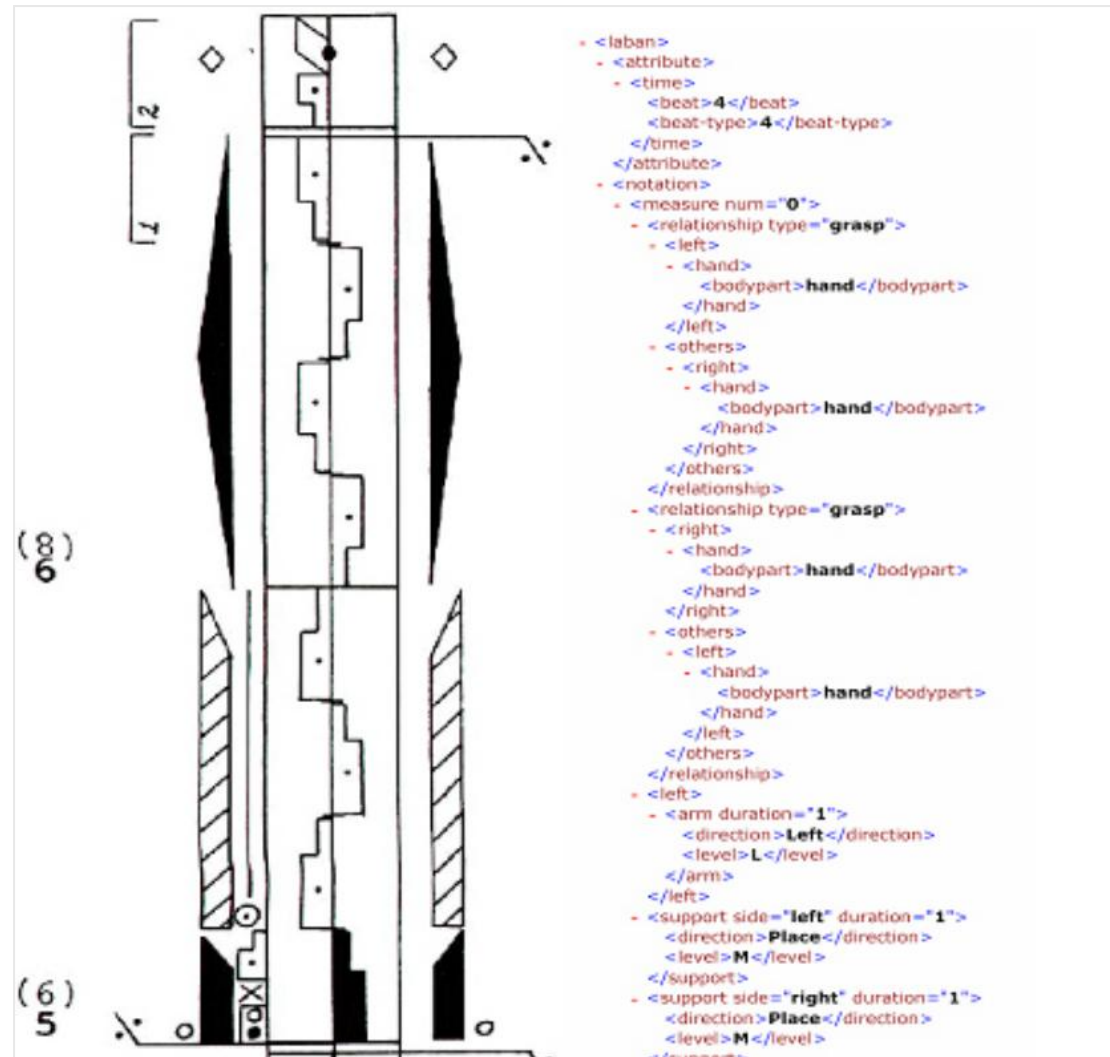
Culturally controlled KG-RAG based Image Description



To what extent can choreographers be supported by semi-automatic dance analysis and the generation of new creative elements in choreographies?



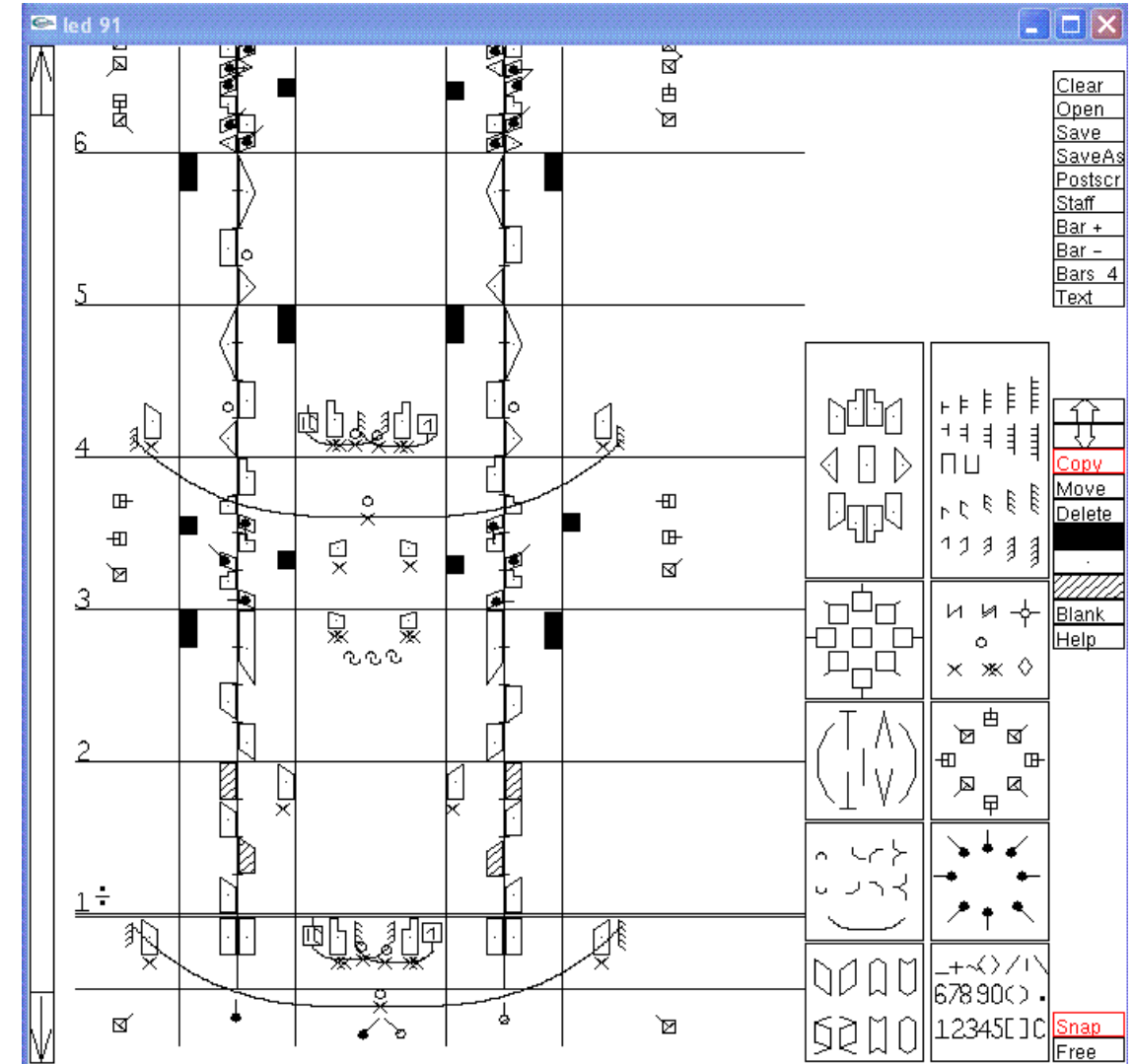
LabanXML and Laban Editor



The image displays a Labanotation score on the left and its corresponding XML representation on the right. The score is a vertical staff with various symbols, including arrows, flags, and beams, indicating movement directions and durations. The XML code on the right is structured as follows:

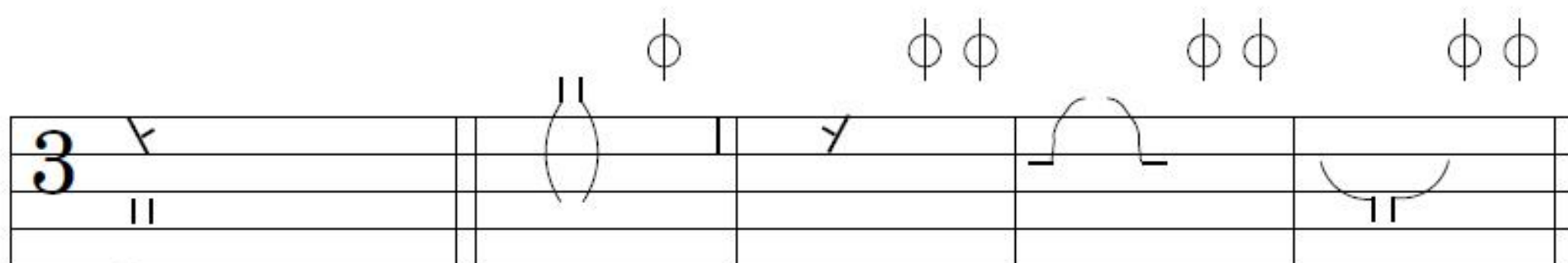
```

<laban>
  <attribute>
    <time>
      <beat>4</beat>
      <beat-type>4</beat-type>
    </time>
  </attribute>
  <notation>
    <measure num="0">
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          <hand>
            <bodypart>hand</bodypart>
          </hand>
        </left>
        <others>
          <right>
            <hand>
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          </right>
        </others>
      </relationship>
      <relationship type="grasp">
        <right>
          <hand>
            <bodypart>hand</bodypart>
          </hand>
        </right>
      </relationship>
      <left>
        <arm duration="1">
          <direction>Left</direction>
          <level>L</level>
        </arm>
      </left>
      <support side="left" duration="1">
        <direction>Place</direction>
        <level>M</level>
      </support>
      <support side="right" duration="1">
        <direction>Place</direction>
        <level>M</level>
      </support>
    </measure>
  </notation>
</laban>
  
```



The image shows a screenshot of the LED Labanotation editor interface. The main window displays a Labanotation score on a grid. The interface includes a menu bar with options like Clear, Open, Save, SaveAs, Postscr, Staff, Bar +, Bar -, Bars 4, and Text. On the right side, there are two panels: one with icons for various Labanotation symbols and another with a list of symbols (Copy, Move, Delete, Blank, Help). The bottom right corner has a 'Snap' button and a 'Free' button.

Benesh



0
0s



36

56



82

100

1s

2s

3s

4s

Ballet languages/systems

Cecchetti system

7 elementary movements: [*plie (bend)*, *etandre (stretch)*, *releve (rise)*, *sauter (jump)*, *glisse (glide)*, *tourne (turn)*, *elancer (dart)*]

Positions: *1st*, *2nd*, *3rd*,... (*left right croise*)

Facing position (*1...8*)

Position in space

Direction of movement (*de cote, dessous, dessus, en avant, en arriere, devant, derriere*)

Combinations (100+) *pas-de-chat, pas-de-bourre, piroutte*



Based on interview with Marije Koning

Dance representation ontology

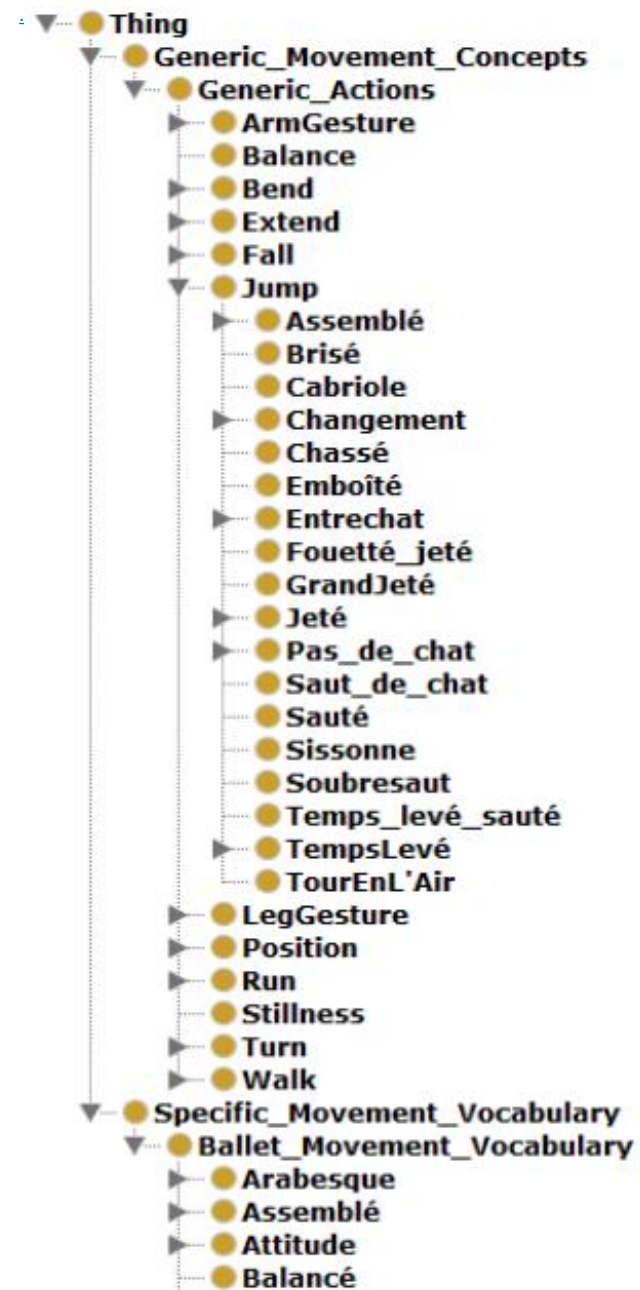


Figure 1-a Protégé view of part of the ontology

El Raheb, et al. **BalOnSe**: Ballet Ontology for Annotating and Searching Video performances. In *Proceedings of the 3rd International Symposium on Movement and Computing* (p. 5). ACM, 2016

Dancepiration – a tool for choreography assistance

Proof-of-concept mobile app

3 different dance styles

Ballet (including 78 steps)

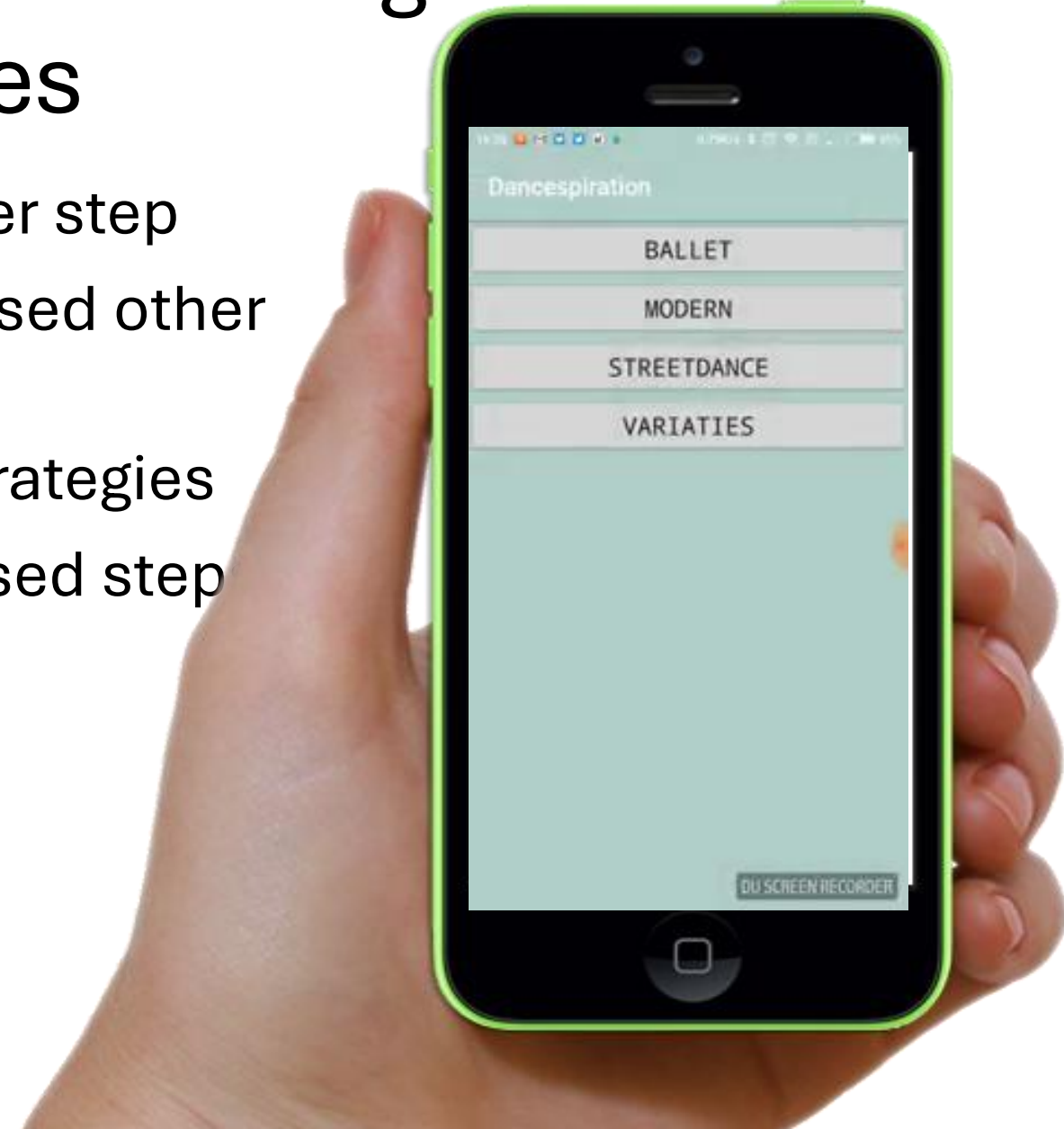
Modern dance (including 57 steps)

Street dance (including 31 steps)



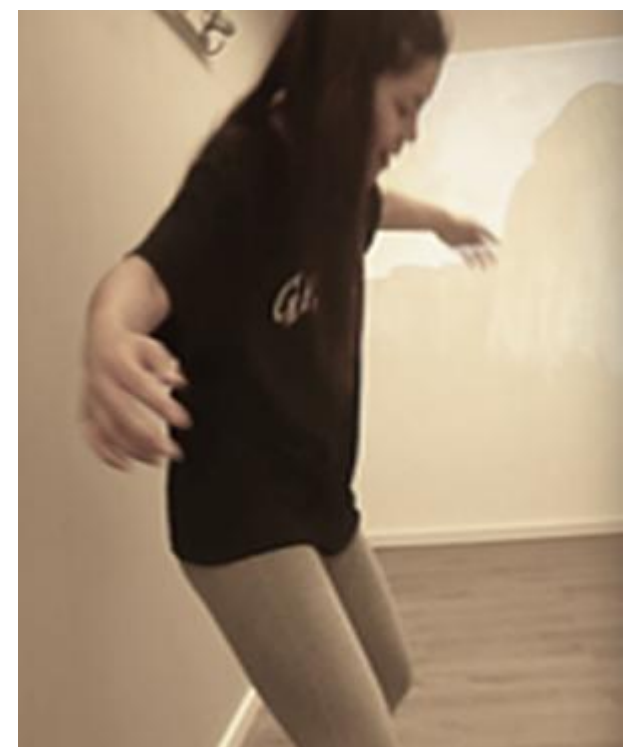
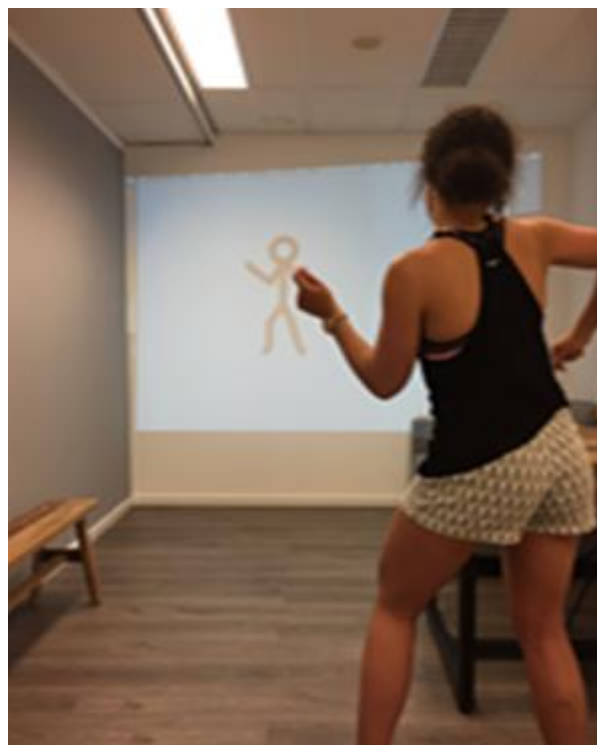
4 rule-based strategies for creating variations on existing choreographies

1. Random step replaced by random other step
2. Random step replaced by ontology-based other step
3. Random steps replaced by multiple strategies
4. Specific step replaced by ontology-based step



4 rule-based strategies for creating variations on existing choreographies

Score	Original	Random	Ontology-Based	Difference	Sig
Average grade	6.17	5.50	6.35	+0.85	**
Correctness		2.89	3.37	+0.48	*
Creativity		3.19	3.37	+0.18	
Helpfulness		2.59	3.00	+0.41	
Meaningfulness		2.70	2.96	+0.26	



Overall assessment per style

Dancehall

vs

Hip-hop

